Chapter 1. Vector Analysis

1. Vector Algebra

1) Triple Products

$\vec{A}\cdot(\vec{B}\times\vec{C}) = \vec{B}\cdot(\vec{C}\times\vec{A}) = \vec{C}\cdot(\vec{A}\times\vec{B})$ (Rotationally preserved order)

Note:

$|\vec{A}\cdot(\vec{B}\times\vec{C})|$ is the volume of the parallelepiped（平行六面体）

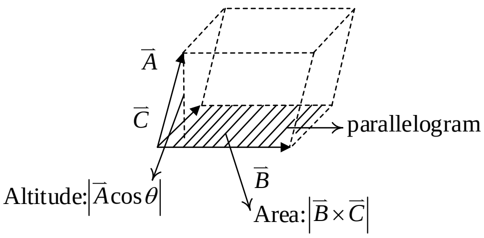

Figure. Geometric interpretation of $|\vec{A}\cdot(\vec{B}\times\vec{C})|$

$\vec{A}\times(\vec{B}\times\vec{C}) = \vec{B}(\vec{A}\cdot\vec{C}) - \vec{C}(\vec{A}\cdot\vec{B})$（bac－cab rule）

（All can be proven using component form）

2) Position, displacement, and separation vectors

$\vec{r} = x\,\hat{x} + y\,\hat{y} + z\,\hat{z}$, $\hat{r} = \frac{x\,\hat{x}+y\,\hat{y}+z\,\hat{z}}{\sqrt{x^2+y^2+z^2}}$

$\vec{\mathscr{r}} = (x-x')\hat{x} + (y-y')\hat{y} + (z-z')\hat{z}$ (separation vector), $\hat{\mathscr{r}} = \frac{\vec{\mathscr{r}}}{\sqrt{(x-x')^2+(y-y')^2+(z-z')^2}}$

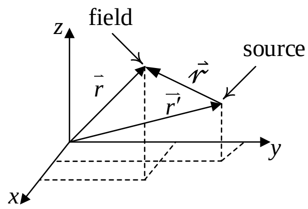

Figure. Definition of field, source, and separation vectors.

$d\vec{l} = dx\,\hat{x} + dy\,\hat{y} + dz\,\hat{z}$ (Infinitesimal displacement vector, sometimes denoted as $d\vec{r}$)

Note: dx, dy, dz don't have to be equal. They determine the direction of $d\vec{r}$.

2. Differential calculus ($\nabla T, \nabla\cdot\vec{v}, \nabla\times\vec{v}$)

1) Gradient:

$$dT = \left(\frac{\partial T}{\partial x}\right)dx + \left(\frac{\partial T}{\partial y}\right)dy + \left(\frac{\partial T}{\partial z}\right)dz$$

$$= (\partial_x T\ \hat{x} + \partial_y T\ \hat{y} + \partial_z T\ \hat{z}) \cdot (dx\,\hat{x} + dy\ \hat{y} + dz\ \hat{z})$$

$$= (\nabla T)\cdot(d\vec{l}),$$

where $\nabla T \equiv (\partial_x T\ \hat{x} + \partial_y T\ \hat{y} + \partial_z T\ \hat{z})$ **operates on a scaler to get a vector**. The direction of ∇T presents the direction of fastest change of T; $|\nabla T|$ denotes the slope of change. Note: $\nabla \equiv (\hat{x}\partial_x + \hat{y}\partial_y + \hat{z}\partial_z)$ is the "Del" operator, which has the following properties:

①A "vector" operator.

②Always operates from the left of a scalar or a vector.

③Mimics an ordinary vector ($\nabla A,\ \nabla\cdot\vec{A},\ \nabla\times\vec{A}$).

Example:

Find ∇r, where $r = \sqrt{x^2+y^2+z^2}$ (cannot include the origin)

Solution: $\nabla r = (\partial_x r\,\hat{x} + \partial_y r\ \hat{y} + \partial_z r\ \hat{z})$

$$= \frac{1}{2}\frac{2x}{r}\hat{x} + \frac{1}{2}\frac{2y}{r}\hat{y} + \frac{1}{2}\frac{2z}{r}\hat{z} = \frac{x\hat{x}+y\hat{y}+z\hat{z}}{r} = \frac{r\hat{r}}{r} = \hat{r}$$

---

2) Divergence ($\vec{A}\cdot\vec{B}$ is a scalar, so is $\nabla\cdot\vec{B}$)

$$\nabla\cdot\vec{v} = (\hat{x}\partial_x + \hat{y}\partial_y + \hat{z}\partial_z)\cdot(v_x\,\hat{x} + v_y\hat{y} + v_z\hat{z}) = \partial_x v_x + \partial_y v_y + \partial_z v_z$$

which is very special because terms such as $\partial_x v_y,\ \partial_y v_x,\ \partial_y v_z,\ \partial_z v_y$ are lacking.

Note:

① $v_x(x,y,z)$ $v_y(x,y,z)$ ... are scalar functions of position (x, y, z).

② The physical meaning of divergence is difficult to understand. But remember, diverge = spread out. Think about a fountain with water flowing out.

③ The divergence itself is a function of position.

Example: $\vec{v} = \frac{\hat{r}}{r^2}$, compute $\nabla\cdot\vec{v}$.

Solution: $\nabla\cdot\vec{v} = (\hat{x}\partial_x + \hat{y}\partial_y + \hat{z}\partial_z)\cdot(\frac{\vec{r}}{r^3}) = (\hat{x}\partial_x + \hat{y}\partial_y + \hat{z}\partial_z)\cdot(\frac{x\,\hat{x}+y\,\hat{y}+z\,\hat{z}}{(x^2+y^2+z^2)^{3/2}})$

$$= \partial_x((x^2+y^2+z^2)^{-\frac{3}{2}}\cdot x) + \partial_y((x^2+y^2+z^2)^{-\frac{3}{2}}\cdot y) + \ \dots\dots$$

$$= \left[-\frac{3}{2}\cdot(x^2+y^2+z^2)^{-\frac{5}{2}}\cdot 2x\cdot x + (x^2+y^2+z^2)^{-\frac{3}{2}}\right] + \ \dots\dots$$

$$= \left[-3(x^2+y^2+z^2)^{-\frac{5}{2}} \cdot x^2 + (x^2+y^2+z^2)^{-\frac{3}{2}}\right] + \dots\dots$$

$$= -3(x^2+y^2+z^2)^{-\frac{5}{2}}(x^2+y^2+z^2) + 3(x^2+y^2+z^2)^{-\frac{3}{2}} = 0$$

This is a weird result because when you plot $\vec{v} = \frac{\hat{r}}{r^2}$, it's highly divergent, as illustrated below.

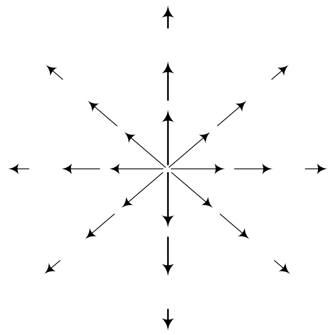

Figure: field distribution of $\vec{v}$.

We will explain this seemingly contradictory result later.

Note: difference between divergence and covariant derivative

$$\vec{\nabla}\vec{A} = (\partial A_i/\partial x_j)_{ij}, \text{lite direct peodoct } \vec{A}\,\vec{B} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} [b_1 \;\; b_2 \;\; b_3]$$

3) Curl:

$$\vec{A} \times \vec{B} = (A_x, A_y, A_z) \times (B_x, B_y, B_z) = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

$$\nabla \times \vec{B} = (\partial_x, \partial_y, \partial_z) \times (B_x, B_y, B_z) = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial_x & \partial_y & \partial_z \\ B_x & B_y & B_z \end{vmatrix}$$

$$= \hat{x}(\partial_y B_z - \partial_z B_y) + \hat{y}(\partial_z B_x - \partial_x B_z) + \hat{z}(\partial_x B_y - \partial_y B_x)$$

Note: the curl of a vector $\vec{A}$ evaluated at point $\vec{P}$ is a vector representing the direction and strength of how $\vec{A}$ swirls around $\vec{P}$.

Example: $\vec{v} = -y\hat{x} + x\hat{y}$, compute $\nabla \times \vec{v}$.

Solution:

$$\nabla \times \vec{v} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial_x & \partial_y & \partial_z \\ -y & x & 0 \end{vmatrix}$$

$$= 0\hat{x} + 0\hat{y} + (\partial_x x - \partial_y(-y))\hat{z} = 2\hat{z}$$

Note: the curl obeys the "right-hand rule", as illustrated in the following figure.

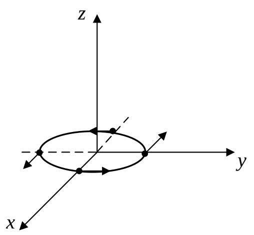

Figure. Illustration of *v*.

---

4) **Product rules**

Recall what we have learnt in basic calculus:

$(f+g)' = f' + g'$, (a)

$(k\,f\,)' = k\,f\,'$, (where k is a constant), (b)

$(f\,g)' = fg' + gf'$, (c)

$\left(\frac{g}{f}\right) = \frac{gf\prime - fg\prime}{g^2}$. (d)

Now let's extend them to vector forms by replacing the derivative with ∇, $\nabla\,\cdot$, and $\nabla\,\times$. Formulas (a), (b) are trivial; however, (c) is NOT trivial:

(i) $\nabla(fg) = f\,\nabla g + g\,\nabla f$ (simple ☺)

(ii) $\nabla(\underbrace{\vec{A}\cdot\vec{B}}_{\text{a scalar}}) = \vec{A}\times(\nabla\times\vec{B}) + \vec{B}\times(\nabla\times\vec{A}) + (\vec{A}\cdot\nabla)\vec{B} + (\vec{B}\times\nabla)\vec{A}$ (hard ☹)

Note:

① To prove (ii), expand both sides using component form, and check term-by-term.

② What is $(A\cdot\nabla)\vec{B}$? This expression looks wrong because ∇ always operates from the left and there is no definition of $\nabla\vec{B}$ (gradient of a vector). In fact,

$$(A\cdot\nabla)\vec{B} = [(A_x, A_y, A_z)\cdot(\partial_x, \partial_y, \partial_z)]\begin{bmatrix} B_x \\ B_y \\ B_z \end{bmatrix} = \underbrace{(A_x\partial_x + A_y\partial_y + A_z\partial_z)}_{\text{a scalar}}\begin{bmatrix} B_x \\ B_y \\ B_z \end{bmatrix}$$

$$= \begin{bmatrix} A_x\partial_x B_x + A_y\partial_y B_y + A_z\partial_z B_z \\ A_x\partial_x B_y + A_y\partial_y B_y + A_z\partial_z B_y \\ A_x\partial_x B_z + A_y\partial_y B_z + A_z\partial_z B_z \end{bmatrix}$$

(iii) $\nabla\cdot(f\vec{A}) = f(\nabla\cdot\vec{A}) + A\cdot(\nabla f)$ (not too bad ☺)

Proof: left: $(\partial_x, \partial_y, \partial_z) \cdot (fA_x, fA_y, fA_z)$

$$= A_x\partial_x f + f\partial_x A_x + A_y\partial_y f + f\partial_y A_y + A_z\partial_z f + fA_z\partial_z$$

Right: $f(\partial_x A_x + \partial_y A_y + \partial_z A_z) + (A_x, A_y, A_z) \cdot (\partial_x f, \partial_y f, \partial_z f)$

$$= A_x\partial_x f + f\partial_x A_x + A_y\partial_y f + f\partial_y A_y + A_z\partial_z f + fA_z\partial_z$$

(iv) $\nabla \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot (\nabla \times \vec{A}) - \vec{A} \cdot (\nabla \times \vec{B})$ (simple ☺)

(v) $\nabla \times (f\vec{A}) = f(\nabla \times \vec{A}) - A \times (\nabla f)$ (simple ☺)

(vi) $\nabla \times (\vec{A} \times \vec{B}) = (\vec{B} \cdot \nabla)\vec{A} - (\vec{A} \cdot \nabla)\vec{B} + \vec{A}(\nabla \cdot \vec{B}) - \vec{B}(\nabla \cdot \vec{A})$ (hard ☹)

(vii) $\nabla\left(\frac{f}{g}\right) = \frac{g\nabla f - f\nabla g}{g^2}, \nabla\left(\frac{\vec{A}}{g}\right) = \frac{g(\nabla \cdot \vec{A}) - \vec{A} \cdot (\nabla g)}{g^2}, \nabla \times \left(\frac{\vec{A}}{g}\right) = \frac{g(\nabla \times \vec{A}) + \vec{A} \times (\nabla g)}{g^2}$

(quotient rules)

5) Second derivatives

(i) $\nabla \cdot (\nabla T) = (\hat{x}\partial_x + \hat{y}\partial_y + \hat{z}\partial_z) \cdot (\partial_x T\hat{x} + \partial_y T\hat{y} + \partial_z T\hat{z})$

$$= \partial_{xx}T + \partial_{yy}T + \partial_{zz}T = \nabla^2 T$$

Note: while ∇ is a 'vector' operator, ∇^2 is a 'scalar' operator.

Exercise: Prove $\nabla^2(\nabla \times \vec{A}) = \nabla \times (\nabla^2\vec{A})$

(ii) $\nabla \times (\nabla T) \equiv 0$

Proof: $\nabla \times (\nabla T) = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial_x & \partial_y & \partial_z \\ \partial_x T & \partial_y T & \partial_z T \end{vmatrix} = 0$

$(\because \partial_{xy} = \partial_{yx}, \partial_{yz} = \partial_{zy}, \partial_{xz} = \partial_{zx})$

(iii) $\nabla \cdot (\nabla \times \vec{v}) \equiv 0$

Proof: $\nabla \cdot (\nabla \times \vec{v})$

$$= (\hat{x}\partial_x + \hat{y}\partial_y + \hat{z}\partial_z) \cdot [\hat{x}(\partial_y v_z - \partial_z v_y) + \hat{y}(\partial_z v_x - \partial_x v_z) + \hat{z}(\partial_x v_y - \partial_y v_x)]$$

$$= \partial_{xy}v_z - \partial_{xz}v_y + \partial_{yz}v_x - \partial_{xy}v_z + \partial_{xz}v_y - \partial_{yz}v_x = 0$$

(iv) $\nabla \times (\nabla \times \vec{v}) = \nabla(\nabla \cdot \vec{v}) - \nabla^2\vec{v}$

Proof : Left $= \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial_x & \partial_y & \partial_z \\ \partial_y v_z - \partial_z v_y & \partial_z v_x - \partial_x v_z & \partial_x v_y - \partial_y v_x \end{vmatrix}$

$$= \begin{bmatrix} \partial_{xy}v_y + \partial_{xz}v_z - \partial_{yy}v_x - \partial_{zz}v_x \\ \partial_{yz}v_z + \partial_{xy}v_x - \partial_{zz}v_y - \partial_{xx}v_y \\ \partial_{xz}v_x + \partial_{yz}v_y - \partial_{xx}v_z - \partial_{yy}v_z \end{bmatrix}$$

$$\text{Right} = \begin{bmatrix} \partial_{xx}v_x + \partial_{xy}v_y + \partial_{xz}v_z \\ \partial_{yx}v_x + \partial_{yy}v_y + \partial_{yz}v_z \\ \partial_{xx}v_x + \partial_{zy}v_y + \partial_{zz}v_z \end{bmatrix} - \begin{bmatrix} \partial_{xx}v_x + \partial_{yy}v_x + \partial_{zz}v_x \\ \partial_{xx}v_y + \partial_{yy}v_y + \partial_{zz}v_y \\ \partial_{xx}v_z + \partial_{yy}v_z + \partial_{zz}v_z \end{bmatrix} = \text{Left}$$

3. Integral calculus

Recall that in basic calculus, $\int_a^b \left(\frac{df}{dx}\right) dx = f(b) - f(a)$, here we extend this formula to line, surface, and volume integrals:

1) The fundamental theorem for gradients

$$dT = (\partial_x T)\, dx + (\partial_y T)\, dy + (\partial_z T)\, dz$$

$$= (\hat{x}\partial_x T + \hat{y}\partial_y T + \hat{z}\partial_z T) \cdot (\hat{x}dx + \hat{y}dy + \hat{z}dz)$$

$$= (\nabla T) \cdot d\vec{l}$$

From the above equation, it is straightforward to prove

$$\int_{\vec{a}}^{\vec{b}} (\nabla T) \cdot d\vec{l} = (\nabla T) \cdot d\overrightarrow{l_1} + (\nabla T) \cdot d\overrightarrow{l_2} + \cdots dT_1 + dT_2 + \cdots + dT_n = T(\vec{b}) - T(\vec{a})$$

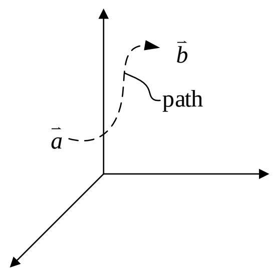

Figure. Integrating from $\vec{a}$ to $\vec{b}$ along a specific path.

Note:

① Line integral $\int_{\vec{a}}^{\vec{b}} (\nabla T) \cdot d\vec{l}$ is independent of the path from $\vec{a}$ to $\vec{b}$.

② $\oint (\nabla T) \cdot d\vec{l} = 0$

Example:

$\vec{v} = \frac{\hat{r}}{r^2}$. Verify that $\int_{\vec{a}}^{\vec{b}} \vec{v} \cdot d\,\vec{l}$ produces the same result regardless of whether it is performed along path ① or ②.

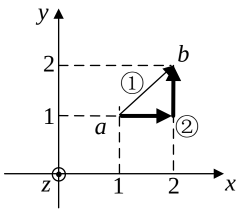

Figure. Integration paths of the example.

Solution:

$\nabla\left(\frac{-1}{r}\right) = \frac{\hat{r}}{r^2} = \vec{v}$, which means $\vec{v}$ is the gradiert of $-\frac{1}{r}$.

So the answer is $\left(-\frac{1}{2\sqrt{2}}\right) - \left(-\frac{1}{\sqrt{2}}\right) = \frac{1}{2\sqrt{2}}$.

To verify, we can integrate along the two paths:

For path 1, $x = y,\ dx = dy,\ z = 0,\ dz = 0$

$$\int_{\vec{a}}^{\vec{b}} \frac{\hat{r}}{r^2} d\vec{l} = \int_{\vec{a}}^{\vec{b}} \frac{x\hat{x} + y\hat{y}}{r^3}(dx\hat{x} + dy\hat{y}) = \int_1^2 \frac{2xdx}{(2x^2)^{3/2}} = \frac{1}{\sqrt{2}}\int_1^2 \frac{dx}{x^2} = \frac{1}{2\sqrt{2}}$$

For path 2, horizontal: $y = 1, dy = 0, z = 0 = dz$

$$\left.\begin{array}{l} \displaystyle\int_1^2 \frac{xdx}{(x^2+1)^{3/2}} = -\int_1^2 d(x^2+1)^{-1/2} = \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{5}} \\ \text{vertioal: } x = 2, dx = 0, z = 0 = dz. \\ \displaystyle\int_1^2 \frac{ydy}{(y^2+4)^{3/2}} = -\int_1^2 d(y^2+4)^{-1/2} = \frac{1}{\sqrt{5}} - \frac{1}{2\sqrt{2}} \end{array}\right\} \int_{path\,2} \frac{\hat{r}}{r^2} d\vec{l} = \frac{1}{2\sqrt{2}}$$

---

2) The fundamental theorem for divergence:

$$\int_V (\nabla\cdot\vec{v})dV = \oint_S \vec{v}\cdot d\vec{a} \left\{\begin{array}{c} \text{Gauss's theorem} \\ \text{Green's theorem} \\ \text{Divergence theorem} \end{array}\right\}$$

where V is volume, S is the outer surface of V, and the surface area element $d\vec{a}$ points outward.

Note:

① The boundary of a line are the two end points, the boundary of a volume is the outer surface.

② Check unit: left: $\frac{[v]\cdot[m^3]}{[m]}$; right: $[v]\cdot[m^2]$.

③ The physical meaning of $\vec{v}\cdot d\vec{a}$ is the total flux of $\vec{v}$ through $d\vec{a}$.

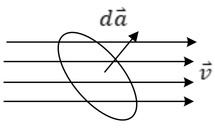

Figure. $\vec{v}\cdot d\vec{a}$ represents the total flux of $\vec{v}$ through $d\vec{a}$

Let's check the theorem with a simple geometry. Consider the volume V enclosed by the cuboidal region shown below:

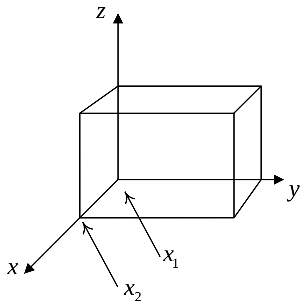

Figure. A cuboidal region in which we verify the divergence theorem.

$$\int_V (\nabla \cdot \vec{v}) dV = \iiint_V (\partial_x v_x + \partial_y v_y + \partial_z v_z) dxdydz = \iint_{y,z} \left[\int_{x_1}^{x_2} \partial_x v_x dx \right] dydz + \cdots$$

$$= \iint_{y,z} v_x(x_2, y, z) dydz - \iint_{y,z} v_x(x_1, y, z) dydz + \cdots$$

$$= \iint_{\substack{\text{Front}\\ \text{surface}}} \underbrace{\vec{v} \cdot d\vec{a}}_{\substack{= v_x \text{dydz.} \\ \text{as } d\vec{a} \text{ points along } \hat{x}}} + \iint_{\substack{\text{Back}\\ \text{surface}}} \underbrace{\vec{v} \cdot d\vec{a}}_{\substack{= v_x \text{dydz.} \\ \text{as } d\vec{a} \text{ points along } -\hat{x}}} + \text{ the rest 4 surfaces}$$

In divergence theorem, $\nabla \cdot \vec{v}$ is the source ("faucet"), $\int \nabla \cdot \vec{v} dV$ is the total fluid generated in the volume. If the fluid is incompressible, the generated fluid must flow out of the region. The total amount flowing out of the outer surface is $\oint \vec{v} \cdot d\vec{a}$.

3) The fundamental theorem for curls:

$$\int_S (\nabla \times \vec{v}) \cdot d\vec{a} = \oint_P \vec{v} \cdot d\vec{l} \quad \text{(Stokes' theorem)}$$

The boundary of an open surface *S* is the perimeter *P*. Now, let's check the theorem with a simple geometry as shown below:

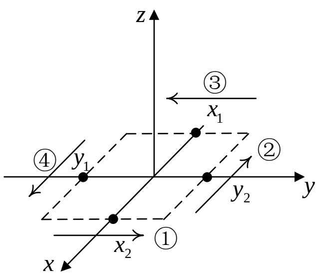

Figure. A rectangular region using which we verify the Stokes' theorem.

$$\oint_p \vec{v}\cdot d\vec{l} = ① + ② + ③ + ④$$

$$= \int_{y_1}^{y_2} v_y(x_2,y)dy - \int_{x_1}^{x_2} v_x(x,y_2)dx - \int_{y_1}^{y_2} v_y(x_1,y)dy + \int_{x_1}^{x_2} v_x(x,y_1)dx$$

$$\int_S (\nabla\times\vec{v})\cdot d\vec{a} = \int_S \{\hat{x}(\partial_y v_z - \partial_z v_y) + \hat{y}(\partial_z v_x - \partial_x v_z) + \hat{z}(\partial_x v_y - \partial_y v_x)\}\cdot \hat{z}dxdy$$

$$= \int_S (\partial_x v_y - \partial_y v_x)\,dxdy$$

$$= \int_{y_1}^{y_2} v_y(x_2,y)dy - \int_{y_1}^{y_2} v_y(x_1,y)dy - \int_{x_1}^{x_2} v_x(x,y_2)dx + \int_{x_1}^{x_2} v_x(x,y_1)dx = \oint_p \vec{v}\cdot d\vec{l}$$

4) Integration by parts

① from product rule: $\nabla\cdot(f\vec{A}) = f(\nabla\cdot\vec{A}) + \vec{A}\cdot(\nabla f)$

$$\Rightarrow \int_V \nabla\cdot(f\vec{A})dV = \int_V f(\nabla\cdot\vec{A})dV + \int_V \vec{A}\cdot(\nabla f)dV = \oint_S f\vec{A}\cdot d\vec{a}$$

$$\Rightarrow \int_V f(\nabla\cdot\vec{A})dV = -\int_V \vec{A}\cdot(\nabla f)dV + \oint_S f\vec{A}\cdot d\vec{a}$$

② from product rule $\nabla\times(f\vec{A}) = f(\nabla\times\vec{A}) - \vec{A}\times(\nabla f)$

$$\Rightarrow \int_S \nabla\times(f\vec{A})\cdot d\vec{a} = \int_S f(\nabla\times\vec{A})\cdot d\vec{a} - \int_S \vec{A}\times(\nabla f)\cdot d\vec{a} = \oint_p f\vec{A}\cdot d\vec{l}$$

$$\Rightarrow \int_S f(\nabla\times\vec{A})\cdot d\vec{a} = \int_S \vec{A}\times(\nabla f)\cdot d\vec{a} + \oint_p f\vec{A}\cdot d\vec{l}$$

③ from product rule $\nabla\cdot(\vec{A}\times\vec{B}) = \vec{B}(\nabla\times\vec{A}) - \vec{A}\cdot(\nabla\times\vec{B})$

$$\Rightarrow \int_V \nabla\cdot(\vec{A}\times\vec{B})\cdot dV = \int_V \vec{B}\cdot(\nabla\times\vec{A})\cdot dV - \int_V \vec{A}\cdot(\nabla\times\vec{B})dV = \oint_S (\vec{A}\times\vec{B})\cdot d\vec{a}$$

$$\Rightarrow \int_V \vec{B}\cdot(\nabla\times\vec{A})dV = \int_V \vec{A}\cdot(\nabla\times\vec{B})dV + \oint_S (\vec{A}\times\vec{B})\cdot d\vec{a}$$

4. The Dirac delta function:

We proved in Section2(2) that $\nabla\cdot\vec{v}$, where $= \frac{\hat{r}}{r^2}$, is equals to 0. Now we take an alternative approach to compute $\nabla\cdot\vec{v}$. In spherical coordinates, the components of $\vec{v}$ can be written as $v_r = \frac{1}{r^2}$, $v_\theta = v_\varphi = 0$.

$$\nabla\cdot\vec{v} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2\cdot v_r) = \frac{1}{r^2}\frac{\partial}{\partial r}(1) = 0$$

Let's integrate $\nabla \cdot \vec{v}$ in a spherical volume, centered at the origin with a radius *R*:

$$\int_V \nabla \cdot \frac{\hat{r}}{r^2} dr = \oint_S \frac{\hat{r}}{r^2} \cdot d\vec{a} = \int_0^{2\pi} \int_0^{\pi} \frac{1}{R^2} \hat{r} \cdot (\hat{r} R^2 sin\theta d\theta d\varphi) = 4\pi$$

$\Rightarrow$ $\nabla \cdot \vec{v}$ cannot be 0 everywhere, otherwise the integration result should be zero.

$\Rightarrow$ $\nabla \cdot \vec{v}$ is 0 everywhere (as proven in Section 2(2)) except at the origin.

The integration including $\vec{r} = 0$ in an arbitrarily small volume is 4π.

$$\Rightarrow \nabla \cdot \frac{\hat{r}}{r^2} = 4\pi\delta^3(\vec{r}), \qquad \nabla \cdot \frac{\hat{\mathscr{r}}}{\mathscr{r}^2} = 4\pi\delta^3(\vec{\mathscr{r}}).$$

In spherical coordinates, $\nabla T = \dfrac{\partial T}{\partial r}\hat{r}$ (T central symmeteic)

$$\Rightarrow \nabla \frac{1}{r} = -\frac{1}{r^2}\hat{r}$$

$$\Rightarrow \nabla \cdot \left(\nabla \frac{1}{r}\right) = \nabla^2 \frac{1}{r} = -4\pi\delta^3(\vec{r}), \qquad \nabla \cdot \left(\nabla \frac{1}{\mathscr{r}}\right) = \nabla^2 \frac{1}{\mathscr{r}} = -4\pi\delta^3(\vec{\mathscr{r}})$$

5. The Helmholtz theorem (*)

Helmholtz's theorem states that a sufficiently smooth vector field can be uniquely determined (up to a constant) if its divergence and curl are known, given appropriate boundary conditions. This theorem is fundamental in electromagnetism, as it ensures that the electric field ($\vec{E}$) and magnetic field ($\vec{B}$) are completely determined by Maxwell's equations.

Can a vector function $\vec{F}$ be completely determined by knowing its divergence and curl?

More specifically, if $\nabla \cdot \vec{F}$ and $\nabla \times \vec{F}$ are given, can we uniquely determine $\vec{F}$?

Solution:

The vector function $\vec{F}(\vec{r})$ can be expressed as $\vec{F}(\vec{r}) = \int_V \vec{F}(\vec{r}')\, \delta^3(\vec{r}' - \vec{r}) dV'$ (sifting property of the delta function)

$\because$ $\delta^3(\vec{r}' - \vec{r}) = -\dfrac{1}{4\pi}\nabla^2 \dfrac{1}{\mathscr{r}}$

$\therefore$ $\vec{F}(\vec{r}) = -\frac{1}{4\pi}\int_V \vec{F}(\vec{r}')\ \nabla^2 \frac{1}{\mathscr{r}}\, dV' = -\frac{1}{4\pi}\nabla^2 \int_v \vec{F}(\vec{r}') \frac{1}{\mathscr{r}} dV' = -\nabla^2 \vec{v}$,

where $\vec{v} = \frac{1}{4\pi}\int_V \vec{F}(\vec{r}') \frac{1}{\mathscr{r}} dV'$

From $\nabla \times \nabla \times \vec{v} = \nabla(\nabla \cdot \vec{v}) - \nabla^2 \vec{v}$

$\vec{F}(\vec{r}) = -\nabla^2 \vec{v} = \nabla \times \nabla \times \vec{v} - \nabla(\nabla \cdot \vec{v})$

$$= \nabla \times \underbrace{\left\{ \nabla \times \left[\frac{1}{4\pi} \int_V \vec{F}(\vec{r}') \frac{1}{\mathscr{r}} dV' \right] \right\}}_{(\mathrm{I})} - \nabla \underbrace{\left\{ \nabla \cdot \left[\frac{1}{4\pi} \int_V \vec{F}(\vec{r}') \frac{1}{\mathscr{r}} dV' \right] \right\}}_{(\mathrm{II})}$$

$$(\mathrm{I}) = \frac{1}{4\pi} \int_V \nabla \times \frac{\vec{F}(\vec{r}')}{\mathscr{r}} dV'$$

$\because \nabla \times \left(\frac{\vec{F}(\vec{r}')}{\mathscr{r}} \right) = \frac{1}{\mathscr{r}} \nabla \times \vec{F}(\vec{r}') - \vec{F}(\vec{r}') \times \left(\nabla \frac{1}{\mathscr{r}} \right)$, note that ∇ operates upon $\vec{r}$ rather than $\vec{r}'$,

so $\nabla \times \vec{F}(\vec{r}') = 0$.

$$\therefore \frac{1}{4\pi} \int_V \nabla \times \frac{\vec{F}(\vec{r}')}{\mathscr{r}} dV' = -\frac{1}{4\pi} \int_v \vec{F}(\vec{r}') \times \nabla \frac{1}{\mathscr{r}} dV'$$

$$= \frac{1}{4\pi} \int_V \vec{F}(\vec{r}') \times \nabla' \frac{1}{\mathscr{r}} dV'$$

(Note: ∇' is the same as ∇ except it operates upon $\vec{r}'$. Since $\mathscr{r} = |\vec{r} - \vec{r}'|$, it is straightforward to prove that $\nabla \frac{1}{\mathscr{r}} = -\nabla' \frac{1}{\mathscr{r}}$).

$\because \nabla' \times \left(\frac{\vec{F}(\vec{r}')}{\mathscr{r}} \right) = \frac{1}{\mathscr{r}} \nabla' \times \vec{F}(\vec{r}') - \vec{F}(\vec{r}') \times \left(\nabla' \frac{1}{\mathscr{r}} \right)$ (Again, this is the product rule, although ∇' operates upon $\vec{r}'$ rather than upon $\vec{r}$.)

$$\therefore \frac{1}{4\pi} \int_V \vec{F}(\vec{r}') \times \nabla' \frac{1}{\mathscr{r}} dV' = \frac{1}{4\pi} \int_V \frac{1}{\mathscr{r}} \nabla' \times \vec{F}(\vec{r}') dV' - \frac{1}{4\pi} \int_v \nabla' \times \left(\frac{\vec{F}(\vec{r}')}{\mathscr{r}} \right) dV'$$

$\because \int_V \nabla \times \vec{B}\ dV = \oint_S d\,\vec{a} \times \vec{B}$ (This is another integral theorem simitar to the divergence theorem)

$$\therefore (\mathrm{I}) = \frac{1}{4\pi} \int_V \nabla \times \frac{\vec{F}(\vec{r}')}{\mathscr{r}} dV' = \frac{1}{4\pi} \int_V \frac{1}{\mathscr{r}} \nabla' \times \vec{F}(\vec{r}') dV' + \frac{1}{4\pi} \oint_s \frac{\vec{F}(\vec{r}')}{\mathscr{r}} \times d\vec{a}'$$

Now let's manipulate (II) in the same manner:

$$(\mathrm{II}) = \frac{1}{4\pi} \int_V \nabla \cdot \frac{\vec{F}(\vec{r}')}{\mathscr{r}} dV'$$

$\because \nabla \cdot (f\vec{A}) = f(\nabla \cdot \vec{A}) + \vec{A} \cdot (\nabla f)$,

here $\nabla \cdot \vec{A}(\vec{r}') = 0$ because ∇ operates on $\vec{r}$ rather than $\vec{r}'$

$$\therefore (\mathrm{II}) = \frac{1}{4\pi} \int_V \vec{F}(\vec{r}') \cdot \left(\nabla \frac{1}{\mathscr{r}} \right) dV' = -\frac{1}{4\pi} \int_V \vec{F}(\vec{r}') \cdot \left(\nabla' \frac{1}{\mathscr{r}} \right) dV'$$

$$= -\frac{1}{4\pi} \int_V \nabla' \cdot \frac{\vec{F}(\vec{r}')}{\mathscr{r}} dV' + \frac{1}{4\pi} \int_V \frac{\nabla' \cdot \vec{F}(\vec{r}')}{\mathscr{r}} dV'$$

$$= \frac{1}{4\pi} \int_V \frac{\nabla' \cdot \vec{F}(\vec{r}')}{\mathscr{r}} dV' - \frac{1}{4\pi} \oint_S \frac{\vec{F}(\vec{r}') \cdot d\vec{a}'}{\mathscr{r}}$$

If $V \to \infty$, $\vec{r}' \to \infty$, $|\vec{r} - \vec{r}'| = \mathscr{r} \approx |\vec{r}'|$

$$\oint_{S\to\infty} \frac{\vec{F}(\vec{r}\,') \times d\vec{a}\,'}{\mathscr{r}} \quad \& \quad \oint_{S\to\infty} \frac{\vec{F}(\vec{r}\,') \cdot d\vec{a}\,'}{\mathscr{r}} \quad \to \text{both approach } \frac{|\vec{F}(\infty)||\vec{r}\,'|^2}{|\vec{r}\,'|} = |\vec{F}(\infty)||\vec{r}\,'|$$

So if $|\vec{F}(\vec{r}\,')| \propto \frac{1}{|\vec{r}\,'|^{1+\epsilon}}, \epsilon > 0,$ both terms approach 0

As a result,

$$(\mathrm{I}) = \underbrace{\frac{1}{4\pi}\int_V \frac{\nabla' \times \vec{F}(\vec{r}\,')}{\mathscr{r}}\, dV'}_{A}, \quad (\mathrm{II}) = \underbrace{\frac{1}{4\pi}\int_V \frac{\nabla' \cdot \vec{F}(\vec{r}\,')}{\mathscr{r}}\, dV'}_{\emptyset}$$

$$\vec{F}(\vec{r}) = \nabla \times (\mathrm{I}) - \nabla(\mathrm{II}) = -\nabla\phi + \nabla \times \vec{A} \qquad (\text{curlless} + \text{divergenceless})$$

where $\phi = \frac{1}{4\pi}\int_V \frac{\nabla' \times \vec{F}(\vec{r}\,')}{\mathscr{r}}\, dV'$, $\vec{A} = \frac{1}{4\pi}\int_V \frac{\nabla' \cdot \vec{F}(\vec{r}\,')}{\mathscr{r}}\, dV'$

Conclusion:

For a vector function $\vec{F}$ satisting $|\vec{F}(\vec{r})| \propto \frac{1}{|\vec{r}|^{1+\epsilon}}$, when r approaches ∞ and $\epsilon > 0$

if $\nabla \times \vec{F}$ *and* $\nabla \cdot \vec{F}$ are given, $\vec{F}$ is uniqely determined by $\vec{F} = -\nabla\phi + \nabla \times \vec{A}$,

where $\phi = \frac{1}{4\pi}\int_V \frac{\nabla' \times \vec{F}(\vec{r}\,')}{\mathscr{r}}\, dV'$, $\vec{A} = \frac{1}{4\pi}\int_V \frac{\nabla' \cdot \vec{F}(\vec{r}\,')}{\mathscr{r}}\, dV'$

Note: $\nabla \times (\nabla\phi) = 0$, meaning that $-\nabla\phi$ is irrotational (curl-less); $\nabla \cdot (\nabla \times \vec{A}) = 0$, meaning that it is solenoidal (divergence-less). As a result, $\vec{F}$ can be split into the sum of an irrotational and a solenoidal component.
